

Spatiotemporal-Filtering-Based Channel Selection for Single-Trial EEG Classification

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Abstract—Achieving high classification performance in electroencephalogram (EEG)-based brain-computer interfaces (BCIs) often entails a large number of channels, which impedes their use in practical applications. Despite the previous efforts, it remains a challenge to determine the optimal subset of channels in a subject-specific manner without heavily compromising the classification performance. In this article, we propose a new method, called spatiotemporal-filtering-based channel selection (STECS), to automatically identify a designated number of discriminative channels by leveraging the spatiotemporal information of the EEG data. In STECS, the channel selection problem is cast under the framework of spatiotemporal filter optimization by incorporating a group sparsity constraints, and a computationally efficient algorithm is developed to solve the optimization problem. The performance of STECS is assessed on three motor imagery EEG datasets. Compared with state-of-the-art spatiotemporal filtering algorithms using full EEG channels, STECS yields comparable classification performance with only half of the channels. Moreover, STECS significantly outperforms the existing channel selection methods. These results suggest that this algorithm holds promise for simplifying BCI setups and facilitating practical utility.

Index Terms—Brain-computer interfaces (BCIs), channel selection, common spatial patterns (CSP), electroencephalogram (EEG), spatiotemporal filtering.

I. INTRODUCTION

BRAIN-COMPUTER interface (BCI) is a promising technology for translating brain signals to external commands [1]–[4]. Due to its relatively low cost and high-temporal resolution, multichannel electroencephalogram (EEG) is generally used in BCI applications [5]–[7]. However, a large number of channels are typically required to achieve

high classification performance, which greatly limits the practicality of BCI systems and their ability to be translated into clinical settings [8], [9]. Finding a minimal number of channels while maintaining the desired classification performance is of tremendous importance for designing a BCI system for daily use.

Many previous studies have addressed the channel selection issue with different strategies, which are briefly reviewed as follows. Lal *et al.* [10] used recursive feature elimination and l_0 -norm optimization within the framework of the support vector machine to selecting channels at the classification stage. Similarly, the Fisher criterion [11] and mutual information [12], [13] were employed to select channels based on the band-power feature. However, these methods treated feature extraction and channel selection as separate stages without considering their mutual influence on each other. Barachant and Bonnet proposed selecting channels in a greedy manner based on backward elimination, with the metric being the Riemann distance of the mean covariance matrices of the two tasks [14]. Nonetheless, greedy algorithms are time consuming and susceptible to local minima.

In more recent years, many studies have attempted to reduce the number of channels based on common spatial patterns (CSP), which is an effective spatial filtering algorithm with a variety of variants [15]–[17]. To select the most discriminative channels, one could select channels by measuring their weight coefficients in the CSP spatial filters [9], [18]. More effectively, channel selection can be achieved by sparsifying the CSP spatial filters. Along this line, Yong *et al.* [19] used a l_1 -norm regularization to enable sparse spatial filters and was able to convert the optimization problem to a quadratically constrained quadratic programming problem, which was solved by sequential quadratic programming and the augmented Lagrangian method. The sparse common spatial pattern (SCSP) [8] replaced the l_1 -norm with the l_1/l_2 -norm, which could be solved similarly as in [19]. The sparse spatial projection (SSP) algorithm [20] exploited an objective function equivalent to CSP's Rayleigh quotient maximization formulation, and formulated channel selection as an unconstrained optimization problem via the l_1 -norm regularization. In addition, greedy approaches, such as forward selection and backward elimination [21], were employed to enable sparse CSP solutions. Binary particle swarm optimization (BPSO) was also combined with CSP to select the best channel subset [22]. These algorithms are typically computationally demanding.

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TABLE I
LIST OF SYMBOLS IN THIS ARTICLE

T : number of sampled time points
C : number of channels
N : order of the optimized temporal filters
k : number of selected channels
$\mathbf{X} \in \mathbb{R}^{C \times T}$: data matrix of single-trial EEG data
$\mathbf{X}^{(\tau)} \in \mathbb{R}^{C \times T}$: data matrix delay $\mathbf{X}$ by τ time steps
$\tilde{\mathbf{X}} \in \mathbb{R}^{N \times C \times T}$: augmented data matrix
$\mathbf{R}_1, \mathbf{R}_2 \in \mathbb{R}^{C \times C}$: estimated covariance matrices of the EEG data under two states
$\hat{\mathbf{R}}_1, \hat{\mathbf{R}}_2 \in \mathbb{R}^{N \times C \times N \times C}$: generalized covariance matrices of the augmented data under two states
$\hat{\mathbf{R}}_1, \hat{\mathbf{R}}_2 \in \mathbb{R}^{N \times K \times N \times K}$: averaged spatio-temporal covariance matrices of the two states after channel selection
$\mathbf{s} \in \mathbb{R}^{C \times 1}$: spatial filter
$\mathbf{a}_j \in \mathbb{R}^{N \times 1}$: temporal filter for the j -th channel
$\mathbf{w} \in \mathbb{R}^{N \times C \times 1}$: combined spatio-temporal filter
$\hat{\mathbf{w}} \in \mathbb{R}^{N \times k \times 1}$: spatio-temporal filters optimized with the selected channels
I_{ch} : index set of selected channels
m : dimension of spatio-temporal feature vectors for each class

Although the above-mentioned channel selection algorithms are able to yield a small number of channels, their classification performance compared with that of CSP was not significantly improved or even declined. In particular, the existing channel selection methods exploited only the spatial information of EEG, while disregarding the potential impact of temporal information on channel selection. Failure to consider both spatial and temporal information likely leads to suboptimal channel selection performance. A number of CSP-based spatiotemporal filtering algorithms have been proposed, such as CSSP [23], CSSSP [24], FBCSP [25], and its variants [26]–[28], SPECCSP [29], DFBCSP [30], ISSPL [31], STRCSP [32], and CFIS [33]. Moreover, we recently proposed the regularized spatiotemporal filtering and classification (RSTFC) [34] algorithm as a new framework for single-trial EEG classification. In view of the superior performance of these spatiotemporal filtering algorithms compared with CSP, channel selection built on these algorithms can potentially achieve better performance.

In this article, an algorithm called spatiotemporal-filtering-based channel selection (STECS) is developed, which aims to select the discriminative channel subset while enhancing the classification performance. The algorithm consists of two steps. First, STECS identifies a designated number of discriminative channels by leveraging the joint spatiotemporal information of the EEG signal. Second, spatiotemporal filtering is performed on the selected channels to improve the classification performance, which is achieved by maximizing the variance ratio of the two tasks, along with regularization to prevent overfitting. In addition, the computational complexity of STECS is modest, and it can be implemented efficiently, which makes the algorithm suitable for quasi-real-time online BCI systems. To demonstrate the effectiveness of STECS, we apply it to three motor imagery datasets, among which two are from past BCI competitions.

The remainder of this article is structured as follows. Section II reviewed the CSP algorithm, followed by the mathematical details of STECS. The performance of STECS was evaluated on the three motor imagery EEG datasets

in Section III, where the details on the datasets, analysis pipeline, and results are provided. Section IV offers more insight into STECS. Finally, Section V presents the conclusions. For ease of reference, Table I provides a list of the essential mathematical symbols used in this article.

II. METHODOLOGY

A. Common Spatial Patterns

As an effective feature extraction method for single-trial EEG classification, CSP optimizes a set of spatial filters to maximally discriminate two mental states [35]–[37]. More specifically, let $\mathbf{X} \in \mathbb{R}^{C \times T}$ be a single-trial EEG data, where C and T denote the numbers of channels and sample points, respectively. CSP seeks to maximize the variance ratio between two mental states

$$\max_{\mathbf{s}} J_1(\mathbf{s}) \triangleq \frac{\mathbf{s}^\top \mathbf{R}_1 \mathbf{s}}{\mathbf{s}^\top \mathbf{R}_2 \mathbf{s}} \quad (1)$$

where $\mathbf{s} \in \mathbb{R}^{C \times 1}$ denotes the spatial filter, $\mathbf{R}_1$ and $\mathbf{R}_2$ are the averaged spatial covariance matrices for the two mental states, and the superscript $\top$ denotes the transpose.

Problem (1) is a Rayleigh quotient maximization problem that can be solved via performing the generalized eigenvalue decomposition of $\mathbf{R}_1$ and $\mathbf{R}_2$

$$\mathbf{R}_1 \mathbf{s} = \lambda \mathbf{R}_2 \mathbf{s} \quad (2)$$

which results in multiple eigenvectors as the spatial filters.

B. STECS

CSP makes use of the EEG from all channels and is unable to perform channel selection by itself. Moreover, the temporal filter is typically determined according to prior neurophysiological knowledge (e.g., a passband of 8–12 Hz or 13–24 Hz is most often used for extracting sensorimotor rhythms in the alpha or beta band), which may not be optimal for each individual subject.

To address these limitations, we propose selecting EEG channels within a spatiotemporal filter optimization framework. In the following text, we first formulate spatiotemporal filter optimization as a Rayleigh quotient maximization problem where the spatial and temporal filters are parameterized collectively, and subsequently show channel selection can be achieved by adding a sparsity-promoting regularization term. Multiple spatial and temporal filters can then be optimized with the selected channels by solving a Rayleigh quotient maximization problem.

1) *Simultaneous Spatial and Temporal Filtering*: In STECS, we assume that the discriminative frequency bands of EEG are channel specific, which is supported by the neurophysiological evidence that the mu and beta rhythms have different spatial topographies in both actual and imagined movements [38]. In light of this assumption, simultaneous spatial and temporal filtering can be accomplished by integrating spatial and temporal filters into a single parameter as described as follows.

Let $\mathbf{s} = [s(1), s(2), \dots, s(C)]^\top \in \mathbb{R}^{C \times 1}$ be a spatial filter, and $\mathbf{a}_j \in \mathbb{R}^{N \times 1}$ be the temporal filter for the j th channel, where N is the order of the temporal filter. Based on our earlier work [34], the spatiotemporally filtered EEG signal can be represented as

$$\tilde{\mathbf{h}} = \mathbf{w}^\top \tilde{\mathbf{X}} \quad (3)$$

where $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_C] \in \mathbb{R}^{N \times C}$, $\mathbf{w} = \text{vec}((\mathbf{s}^\top \odot \mathbf{A})^\top) = [s(1) \cdot \mathbf{a}_1^\top, s(2) \cdot \mathbf{a}_2^\top, \dots, s(C) \cdot \mathbf{a}_C^\top]^\top \in \mathbb{R}^{NC \times 1}$ is the reparametrized spatiotemporal filter, and $\tilde{\mathbf{X}}$ denotes the augmented data matrix

$$\tilde{\mathbf{X}} = \begin{pmatrix} \mathbf{X} \\ \mathbf{X}^{(1)} \\ \vdots \\ \mathbf{X}^{(N-1)} \end{pmatrix} \quad (4)$$

where $\mathbf{X}^{(n)}$, $n \in \{1, 2, \dots, N-1\}$ is the n time step delayed data matrix.

Since $\mathbf{w}$ encodes both the spatial and temporal filters, it can be decomposed into the spatial filter $\mathbf{s}$ and temporal filters $\mathbf{A}$ once the optimal $\mathbf{w}$ is determined. To see this, we partition the spatiotemporal filter $\mathbf{w}$ evenly into C groups: $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_C$, where $\mathbf{w}_c = s(c) \cdot \mathbf{a}_c$, and $s(c)$ is the c th element of the spatial filter $\mathbf{s}$. Assuming each temporal filter $\mathbf{a}_c$ has a unit l_2 -norm, and the sign of the corresponding spatial filter weight $s(c)$ is identical to that of w_{1c} , the spatial filter weight for the c th channel is

$$s(c) = \text{sign}(w_{1c}) \cdot \|\mathbf{w}_c\|_2 \quad (5)$$

where $\|\cdot\|_2$ is the l_2 -norm.

The spatiotemporally filtered EEG signal $\tilde{\mathbf{h}}$ is a linear transform of the augmented data matrix $\tilde{\mathbf{X}}$. Hence, the spatiotemporal filter $\mathbf{w}$ can again be optimized by solving the following variance ratio maximization problem as in CSP:

$$\max_{\mathbf{w}} J_2(\mathbf{w}) \triangleq \frac{\mathbf{w}^\top \tilde{\mathbf{R}}_1 \mathbf{w}}{\mathbf{w}^\top \tilde{\mathbf{R}}_2 \mathbf{w}} \quad (6)$$

where $\tilde{\mathbf{R}}_1$ and $\tilde{\mathbf{R}}_2$ denote the spatiotemporal covariance matrices under the two mental states estimated by the augmented data matrices $\tilde{\mathbf{X}}$'s as in (4).

Intuitively, if the EEG signal of the c th channel does not contain discriminative information about the two mental states, the magnitude of the elements of $\mathbf{w}_c$ should be tiny such that the c th channel can essentially be pruned without affecting the classification performance. This serves as the basis for our channel selection procedure described next.

2) *Channel Selection via Sparse Learning*: To determine the unimportant channels, the corresponding elements in $\mathbf{w}$ should approach zeros. In machine learning, this can be achieved by group-sparsity regularization based on the $l_{2,1}$ -norm of $\mathbf{w}$ defined as $\|\mathbf{w}\|_{2,1} = \sum_{c=1}^C \|\mathbf{w}_c\|_2$ [39]. However, the Rayleigh quotient in problem (6) is scale invariant with respect to the spatiotemporal filter. That is, for a spatiotemporal filter $\mathbf{w}$ and a nonzero real number a

$$J_2(\mathbf{w}) = \frac{\mathbf{w}^\top \tilde{\mathbf{R}}_1 \mathbf{w}}{\mathbf{w}^\top \tilde{\mathbf{R}}_2 \mathbf{w}} = \frac{(a\mathbf{w})^\top \tilde{\mathbf{R}}_1 (a\mathbf{w})}{(a\mathbf{w})^\top \tilde{\mathbf{R}}_2 (a\mathbf{w})} = J_2(a\mathbf{w}). \quad (7)$$

As a result, the group-sparsity regularization term cannot be added directly to the Rayleigh quotient problem, as the objective function can be made arbitrarily large by rescaling the spatiotemporal filter.

Alternatively, solving the following optimization problem yields an identical solution to that of problem (6) (see Appendix A) [20]:

$$\min_{\mathbf{w}} J_3(\mathbf{w}) \triangleq \mathbf{w}^\top \tilde{\mathbf{R}}_2 \mathbf{w} + \frac{1}{\mathbf{w}^\top \tilde{\mathbf{R}}_1 \mathbf{w}}. \quad (8)$$

J_3 is desirable as it is no longer scale invariant to $\mathbf{w}$. As such, the group-sparsity term $\|\mathbf{w}\|_{2,1}$ can be directly incorporated to enable sparse solutions.

Consequently, with the penalty term $\|\mathbf{w}\|_{2,1}$ incorporated in (8), STECS solves the following optimization problem:

$$\min_{\mathbf{w}} J_4(\mathbf{w}) \triangleq \mathbf{w}^\top \tilde{\mathbf{R}}_2 \mathbf{w} + \frac{1}{\mathbf{w}^\top \tilde{\mathbf{R}}_1 \mathbf{w}} + \rho_1 \|\mathbf{w}\|_{2,1} \quad (9)$$

where ρ_1 is the regularization parameter that controls the cardinality of the selected channels. An algorithm for solving problem (9) is presented in Appendix B.

Solving problem (9) yields the optimal spatiotemporal filter $\mathbf{w}_1$ for the 1st mental state, while exchanging $\tilde{\mathbf{R}}_1$ and $\tilde{\mathbf{R}}_2$ in (9) allows us to obtain the optimal spatiotemporal filter $\mathbf{w}_2$ for the 2nd mental state. The spatial filters $\mathbf{s}_1$ and $\mathbf{s}_2$ can then be calculated according to (5), and then channels associated with near-zero weights in both $\mathbf{s}_1$ and $\mathbf{s}_2$ are redundant and should be removed.

3) *Learning Multiple Spatiotemporal Filters*: With the selected channels, we use a nonsparse spatiotemporal filter optimization algorithm to learn multiple spatiotemporal filters, as the use of multiple spatiotemporal filters has been indicated to further increase the classification performance [35].

Let $\hat{\mathbf{R}}_1, \hat{\mathbf{R}}_2 \in \mathbb{R}^{NK \times NK}$ denote the averaged spatiotemporal covariance matrices of the two states after channel selection. With a l_2 -norm regularization added to ameliorate overfitting, the following Rayleigh quotient maximization problem is solved to yield multiple spatiotemporal filters:

$$\max_{\hat{\mathbf{w}}} J_5(\hat{\mathbf{w}}) \triangleq \frac{\hat{\mathbf{w}}^\top \hat{\mathbf{R}}_1 \hat{\mathbf{w}}}{\hat{\mathbf{w}}^\top (\hat{\mathbf{R}}_2 + \rho_2 \cdot \mathbf{I}) \hat{\mathbf{w}}} \quad (10)$$

where ρ_2 is the regularization parameter, and $\rho_2 \cdot \mathbf{I}$ regularizes the spatiotemporal covariance matrix of the 2nd state. The resulting spatiotemporal filters are the eigenvectors corresponding to the m largest eigenvalues. By solving the problem with $\hat{\mathbf{R}}_1$ and $\hat{\mathbf{R}}_2$ exchanged in (10), the m spatiotemporal filters of the other state can be obtained. STECS is summarized in Algorithm 1.

III. EXPERIMENTAL VALIDATION

In this section, we assess the performance of STECS on three motor imagery BCI datasets, including two publicly available datasets from past BCI competitions and one in-house dataset. In particular, STECS is compared with SSP [20], the algorithm proposed in [18] (called CSP-R in the following), and FS and BE [21]. The relative advantages of STECS over other channel selection algorithms are illustrated

Algorithm 1 STECS

Require: multichannel EEG data $\mathbf{X}$ recorded under two experimental states

Ensure: the index set of selected channels denoted as I_{ch} , and the spatiotemporal filters $\hat{\mathbf{w}}_i, i = \{1, 2, \dots, 2m\}$

- 1: Calculate $\hat{\mathbf{R}}_1$ and $\hat{\mathbf{R}}_2$ by $\hat{\mathbf{X}}$'s in (4).
- 2: Optimize $\mathbf{w}_1$ by (9) (and $\mathbf{w}_2$ with $\hat{\mathbf{R}}_1$ and $\hat{\mathbf{R}}_2$ exchanged).
- 3: Decompose $\mathbf{w}_1$ and $\mathbf{w}_2$ into $\mathbf{s}_1, \mathbf{A}_1$ and $\mathbf{s}_2, \mathbf{A}_2$.
- 4: Determine I_{ch} by $\mathbf{s}_1$ and $\mathbf{s}_2$.
- 5: Construct $\hat{\mathbf{R}}_1$ and $\hat{\mathbf{R}}_2$.
- 6: Optimize the spatiotemporal filters $\hat{\mathbf{w}}_i$'s by (10), $i = \{1, 2, \dots, m\}$ [$\hat{\mathbf{w}}_j$'s, $j = \{m+1, \dots, 2m\}$ for another state with $\hat{\mathbf{R}}_1$ and $\hat{\mathbf{R}}_2$ exchanged in (10)].

with intuitive examples. Moreover, we compare STECS with the standard CSP and full-cap-based spatiotemporal filtering methods, including STRCSP [32], CSSP [23], CSSSP [24], and FBCSP [25]. The MATLAB code of STECS is available upon requests from the authors.

A. EEG Data Description

1) *Dataset A (Dataset IVa From BCI Competition III [40]):* The experiment is a classical cue-based motor imagery paradigm with five subjects performed right hand and foot motor imagery tasks (labeled 1, 2). EEG signals were recorded using 118 channels, with a sampling frequency of 100 Hz. A total of 280 trials were available for each subject, among which 224, 168, 84, 56, and 28 trials composed the training set for subjects *al*, *aa*, *av*, *aw*, and *ay*, and the remaining trials composed the testing set.

2) *Dataset B (Dataset IIIa From BCI Competition III [41]):* This dataset contains EEG signals from three subjects (*K3*, *K6*, *L1*) who performed left hand, right hand, foot, and tongue motor imagery tasks (labeled 1, 2, 3, 4). EEG signals were recorded using 60 channels, with a sampling frequency of 250 Hz. Both the training set and the testing set contain 45 trials per class for *K3*, and 30 trials for *K6* and *L1*. For the purpose of binary classification, we split the data of each subject to generate $C_4^2 = 6$ subsets, resulting in 18 data subsets. For convenience, we use “subject’s name-labels” to denote the subsets.

3) *Dataset C:* This in-house dataset was provided by Tsinghua University. Seven healthy volunteers (*S1*–*S7*) participated in this motor imagery experiment, where the tasks were right hand and foot motor imagery. This dataset contains data from several sessions without feedback. EEG signals were recorded using 32 channels and were digitized at 256 Hz. For each subject, the training set was composed of 60 trials (30 trials per class) recorded in a single session. The testing dataset consisted of 240 trials for subjects *S1*–*S4* and 180 trials for subjects *S5*–*S7*.

In total, $5 + 6 \times 3 + 7 = 30$ data subsets were obtained for evaluation. These three datasets are well suited for studying the issue of channel selection, for which the channel layouts are presented in Fig. 1.

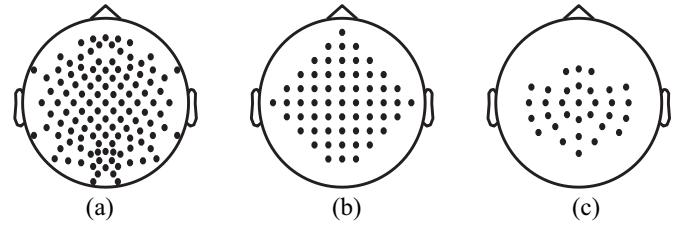


Fig. 1. Channel layouts for a) Dataset A with 118 channels, b) Dataset B with 60 channels, and c) Dataset C with 32 channels.

B. Analysis Pipeline

The workflow of STECS for binary classification is described as follows.

1) *Preprocessing:* The following preprocessing steps are applied to the above-mentioned 30 data subsets.

1) *Channels for Analysis:* We use all of the channels for data analysis and preprocessing, without channel preselection.

2) *Bandpass Filtering:* We use a sixth-order Butterworth filter with the pass-band range of 7–30 Hz to filter the EEG signals, so that the components unrelated to the sensorimotor rhythm can be filtered out.

3) *Time Segments:* The time window is set to 0.5–3.5 s relative to the time point when the visual cue ends.

2) *Channel Selection:* The specific procedures of STECS for channel selection are presented in Sections II-B.

Note that in order to eliminate the impact of the scale of the spatial filters on channel selection, we normalize each spatial filter by the maximum absolute value of its elements

$$\bar{\mathbf{s}}_i = \mathbf{s}_i / \max\{|\mathbf{s}_i(1)|, \dots, |\mathbf{s}_i(C)|\}, i = 1, 2 \quad (11)$$

and the channels in which the absolute values of the corresponding weights do not exceed a predefined threshold 0.01 were discarded.

3) *Feature Extraction:* Multiple spatiotemporal filters are optimized as described in Sections II-B3. To capture the ERD/ERS in the motor imagery-based BCI system, we employ the variance of the spatiotemporally filtered EEG signal as the feature.

Following previous studies, we construct features from the filters corresponding to the three largest generalized eigenvalues (that is, $2m = 6$) for each state [35]. Concretely, let $\mathbf{a} = [a_1, a_2, \dots, a_6]$ denote the feature vector, where a_j is the variance of the EEG signal filtered by the j th filter, $j = \{1, 2, \dots, 6\}$. To render the features more Gaussian distributed, which is the underlying assumption of many linear classifiers, we take the logarithm of the normalized variance of $\mathbf{a}$ as the ultimate features, that is, $\mathbf{z} = [z_1, z_2, \dots, z_6]$, where $z_j = \log(a_j / \sum_{i=1}^6 a_i)$.

4) *Classification:* Fisher’s linear discriminant analysis (FLDA) is used as the classifier due to its low complexity and its effectiveness in single-trial EEG classification [35].

For the other channel selection algorithms, the same preprocessing is employed. Channel selection is then carried out, followed by optimizing multiple spatial filters based on the selected channels. To be consistent with STECS, three spatial

TABLE II
MEAN CLASSIFICATION PERFORMANCE (%) OF STECS, CSP-R, SSP, FS, AND BE ON THE THREE DATASETS

Mean classification performance (%) of each channel selection algorithm on Data Set A.					
k	STECS	CSP-R	SSP	FS	BE
10	78.77 ± 14.66	78.68 ± 16.83	75.06 ± 14.43	66.78 ± 12.00	67.70 ± 13.37
20	82.46 ± 16.34	81.45 ± 13.30	71.88 ± 18.38	70.23 ± 13.24	66.26 ± 16.29
30	78.91 ± 14.09	79.60 ± 12.85	69.93 ± 19.99	71.50 ± 12.57	68.19 ± 16.10
40	80.61 ± 15.25	80.22 ± 12.15	73.36 ± 14.36	73.10 ± 12.09	68.36 ± 14.81
50	81.09 ± 13.86	71.81 ± 17.96	75.50 ± 11.55	72.15 ± 12.33	72.25 ± 12.96
60	80.59 ± 14.52	70.15 ± 18.65	76.84 ± 14.21	76.31 ± 13.40	70.74 ± 14.21
70	79.79 ± 16.81	72.61 ± 17.27	77.31 ± 13.97	75.83 ± 13.15	75.77 ± 14.55
mean	80.32	76.36	74.23	72.27	69.90
Mean classification performance (%) of each channel selection algorithm on Data Set B.					
k	STECS	CSP-R	SSP	FS	BE
10	87.35 ± 8.581	87.96 ± 13.64	85.00 ± 10.79	84.81 ± 9.392	78.95 ± 13.48
20	90.34 ± 9.392	85.09 ± 14.38	88.46 ± 11.27	87.44 ± 9.851	85.52 ± 12.35
30	92.62 ± 7.081	84.29 ± 17.17	88.86 ± 10.70	87.44 ± 9.889	87.22 ± 10.71
40	92.47 ± 7.190	85.22 ± 13.84	88.73 ± 11.45	89.35 ± 9.882	86.57 ± 10.89
mean	90.70	86.64	87.76	87.26	84.57
Mean classification performance (%) of each channel selection algorithm on Data Set C.					
k	STECS	CSP-R	SSP	FS	BE
10	87.62 ± 11.09	84.92 ± 14.91	84.11 ± 15.52	84.36 ± 16.01	84.42 ± 15.98
20	88.06 ± 11.93	86.79 ± 13.77	85.57 ± 11.94	86.23 ± 13.96	85.99 ± 9.731
30	89.80 ± 9.848	83.79 ± 17.06	84.62 ± 15.39	84.36 ± 16.01	84.42 ± 15.98
mean	88.49	85.17	84.77	84.99	84.95

filters corresponding to the largest eigenvalues are obtained for each class. Finally, the power of the spatially filtered signal is used as the feature for classification.

5) *Hyperparameter Determination*: Several hyperparameters need to be determined for STECS. We empirically set the order of the temporal filter of STECS to $N = 6$, and the regularization parameter in problem (10) to $\rho_2 = 10^{-5}$, based on our previous work [32], [34]. To compare the performance of all channel selection methods under each designated number of selected channels, namely, $k \in \{10, 20, 30, 40, 50, 60, 70\}$ for Dataset A, $k \in \{10, 20, 30, 40\}$ for Dataset B, and $k \in \{10, 20, 30\}$ for Dataset C, ρ_1 is determined from a candidate set $B = \{0.01, 0.02, \dots, 0.7\}$ such that the resulting channel number k^* satisfies $||3k^* - k||_1 \leq 1$.

C. Results

1) *Comparisons With Other Channel Selection Algorithms*: Table II compares the classification performance (mean $\pm$ standard deviation) of STECS with that of other channel selection methods on the three datasets. For each algorithm, the classification accuracies associated with different values of k are provided, with the highest entry among the compared algorithms highlighted in boldface for each k . It can be observed that STECS consistently outperforms all other channel selection algorithms, except for Dataset IVa when $k = 30$ and Dataset IIIa when $k = 10$, where STECS performs slightly worse than CSP-R ($p > 0.05$).

To assess the overall performance of each algorithm, the mean classification accuracies with the standard error of the mean (s.e.m.) over all the data subsets are presented in Fig. 2. Note that Dataset C has only 32 channels and thus is not considered for $k = 40$. For each k , the statistical significance of each algorithm versus STECS is assessed via Wilcoxon signed-rank test. In terms of the mean classification accuracies,

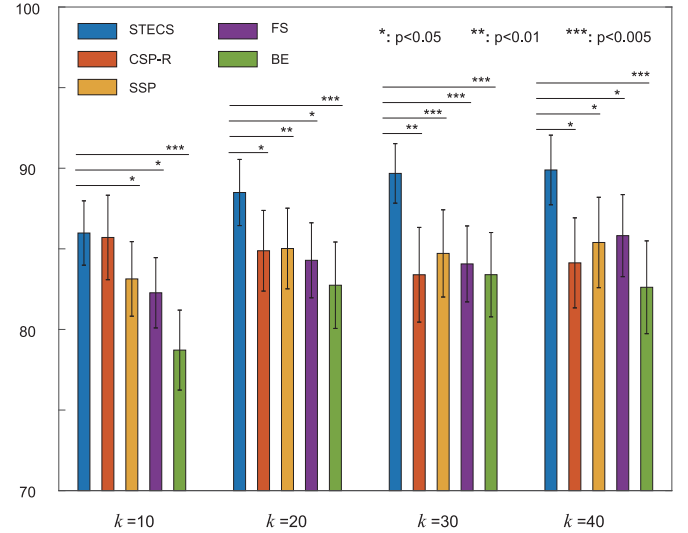


Fig. 2. Mean classification performance (%) averaged across all considered subjects of each algorithm (mean $\pm$ s.e.m.). The black asterisk * above certain bars denotes that the performance of STECS is significantly better than that of the corresponding algorithm at the 0.05 level, and ** and *** indicate $p < 0.01$ and $p < 0.005$, respectively.

STECS dominates the compared algorithms. The differences are significant except for STECS versus CSP-R when $k = 10$.

It can also be observed that the performance improvement by using STECS is notable for some of the subjects poorly classified by CSP-R, SSP, FS, and BE, for example, subsets *aa*, *aw*, *K6-14*, *K6-24*, *L1-34*, and *S3*. The testing classification accuracies per subject associated with different values of k are presented in Table III, and the highest entry among the compared algorithms is highlighted in boldface. Subjects (e.g., *al*, *K3*, *S6*, and *S7*) with already excellent accuracies by other channel selection methods exhibit no particular advantage of using STECS since there is little room for improvement.

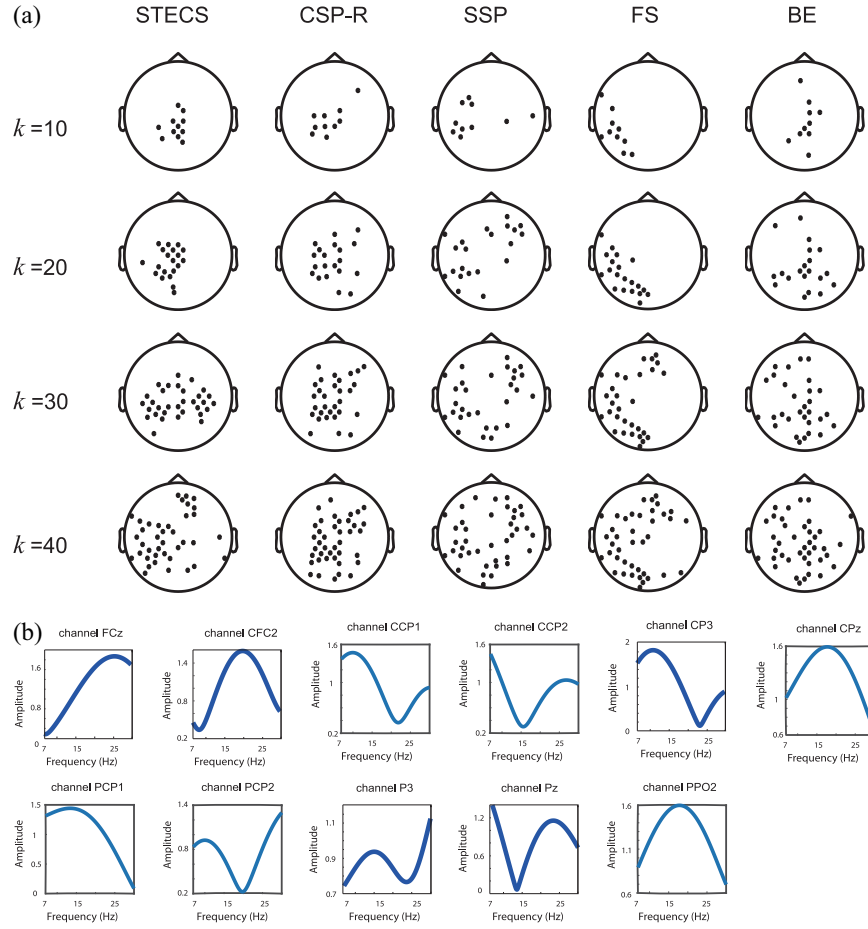


Fig. 3. (a) Topographical maps of the selected channels for subject *aw*, for which the performing tasks were right-hand and foot motor imagery. (b) Frequency response curves for the optimal temporal filters from subject *aw*, and the eleven channels are the selected when $k = 10$.

TABLE III
CLASSIFICATION PERFORMANCE (%) OF STECS, CSP-R, SSP, FS, AND BE ON SUBSETS *aa*, *aw*, *K6-14*, *K6-24*, *L1-34*, AND *S3*

$k = 10$	STECS	CSP-R	SSP	FS	BE	$k = 20$	STECS	CSP-R	SSP	FS	BE
<i>aa</i>	68.75	66.07	68.75	59.82	51.79	<i>aa</i>	75.89	64.29	67.86	58.93	52.68
<i>aw</i>	92.41	74.55	70.98	55.36	78.57	<i>aw</i>	91.07	90.18	54.02	58.93	55.36
<i>K6-14</i>	95.00	91.67	88.33	78.33	85.00	<i>K6-14</i>	98.33	50.00	95.00	88.33	88.33
<i>K6-24</i>	83.33	50.00	80.00	76.67	80.00	<i>K6-24</i>	85.00	73.33	78.33	73.33	81.67
<i>L1-34</i>	83.33	73.33	66.67	80.00	75.00	<i>L1-34</i>	80.00	76.67	66.67	70.00	70.00
<i>S3</i>	84.17	82.92	75.00	81.67	82.08	<i>S3</i>	80.83	80.42	77.08	74.17	75.83
$k = 30$	STECS	CSP-R	SSP	FS	BE	$k = 40$	STECS	CSP-R	SSP	FS	BE
<i>aa</i>	83.93	65.18	65.18	60.71	55.36	<i>aa</i>	84.82	66.96	64.29	58.93	62.50
<i>aw</i>	80.36	76.34	48.66	66.52	58.93	<i>aw</i>	90.63	83.93	61.61	73.21	56.25
<i>K6-14</i>	98.33	93.33	95.00	76.67	91.67	<i>K6-14</i>	98.33	93.33	96.67	85.00	91.67
<i>K6-24</i>	90.00	83.33	76.67	75.00	83.33	<i>K6-24</i>	88.33	78.33	75.00	78.33	83.33
<i>L1-34</i>	83.33	66.67	65.00	75.00	73.33	<i>L1-34</i>	81.67	78.33	73.33	80.00	78.33
<i>S3</i>	84.17	70.00	70.42	68.75	70.00						

2) *Comparisons With Full-Cap-Based Algorithms:* As our goal is to reduce the number of channels while maintaining high classification performance, we next compare the performance of STECS with that of full-cap-based CSP and STRCSP, the latter of which is a full-cap-based spatiotemporal filtering method we previously developed [32], [34].

The overall mean classification accuracy of CSP on the three datasets is $86.03\% \pm 13.05\%$, with the accuracies on the three datasets being $75.44\% \pm 14.83\%$, $89.78\% \pm 9.46\%$, and $86.67\% \pm 12.07\%$, respectively. The performance of CSP

is significantly lower than that of STECS when $k \geq 20$ (see Tables I and II in Appendix C). When $k = 10$, the mean classification accuracies of STECS are not higher than those of CSP, though the differences are not significant. Therefore, despite using fewer channels, STECS outperforms the full-cap-based CSP in most cases.

In line with the settings of STECS, the hyperparameters in STRCSP are set to $N = 6$ and $\rho_2 = 10^{-5}$. The overall mean classification accuracy of STRCSP on the three datasets is $88.84\% \pm 11.71\%$, with the accuracy on the three

TABLE IV
TOXONOMY OF CHANNEL SELECTION AND FEATURE EXTRACTION ALGORITHMS

Algorithms	Channel Selection			Feature Extraction	
	Spatial Filtering	Temporal Filtering	No Channel Selection	Spatial Filtering	Temporal Filtering
STECS	✓	✓		✓	✓
CSP			✓	✓	
CSP-R, SSP, FS/BE	✓			✓	
STRCSP, CSSP, CSSSP, FBCSP			✓	✓	✓

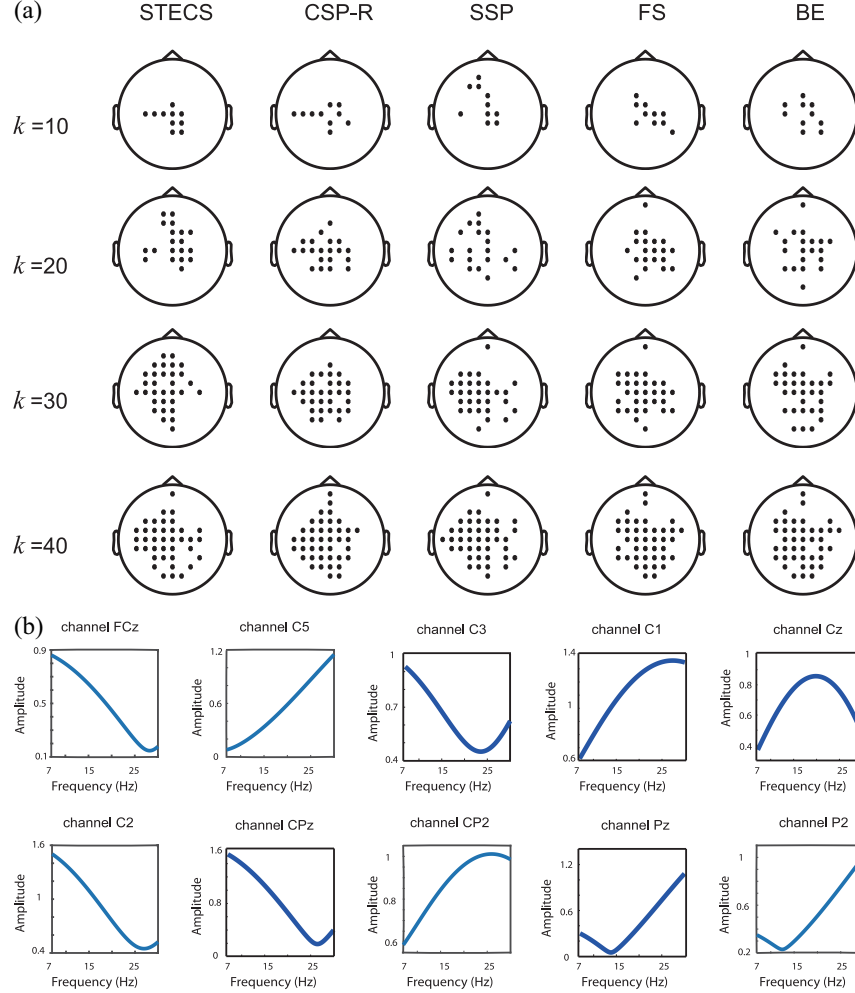


Fig. 4. (a) Topographical maps of the selected channels for subject K3-23, for which the performing tasks were right-hand and foot motor imagery. (b) Frequency response curves for the optimal temporal filters from subject K3-23, and the ten channels are selected when $k = 10$.

datasets being $80.20\% \pm 16.89\%$, $91.89\% \pm 8.061\%$, and $89.35\% \pm 11.22\%$, respectively. When $k = 10$, the performance of STRCSP is significantly better than that of STECS (the mean classification accuracies of STRCSP exceed those of STECS by approximately 3%). Despite that there is no significant performance difference between STECS and STRCSP when $k = 20, 30$, and 40 , the mean classification accuracies of STECS are higher than those of STRCSP when $k = 30, 40$.

Appendix C provides performance comparisons of STECS with more spatiotemporal filtering methods, including CSSP, CSSSP, and FBCSP, where STECS consistently outperforms these methods.

In summary, the above results demonstrate that STECS significantly outperforms other channel selection methods, and achieves performance comparable to or even better than that

of CSP and STRCSP when the number of selected channels is as low as 20.

3) *Visualizing Selected Channels*: To ensure that the selected channels are neurophysiologically meaningful, we provide representative examples to visualize their patterns. For subjects *aw*, K3-23, and S7, where the motor imagery tasks are right-hand versus foot, the topographical maps of the channels selected by each algorithm (STECS, CSP-R, SSP, FS, and BE) are shown in Figs. 3–5, respectively, along with the frequency response curves of the optimal temporal filters for all the selected channels when $k = 10$.

The channels selected by STECS are predominantly centered at the contralateral sensorimotor and posterior parietal regions, which are in accordance with the brain areas where ERD is expected to appear for right-hand and foot imagination,

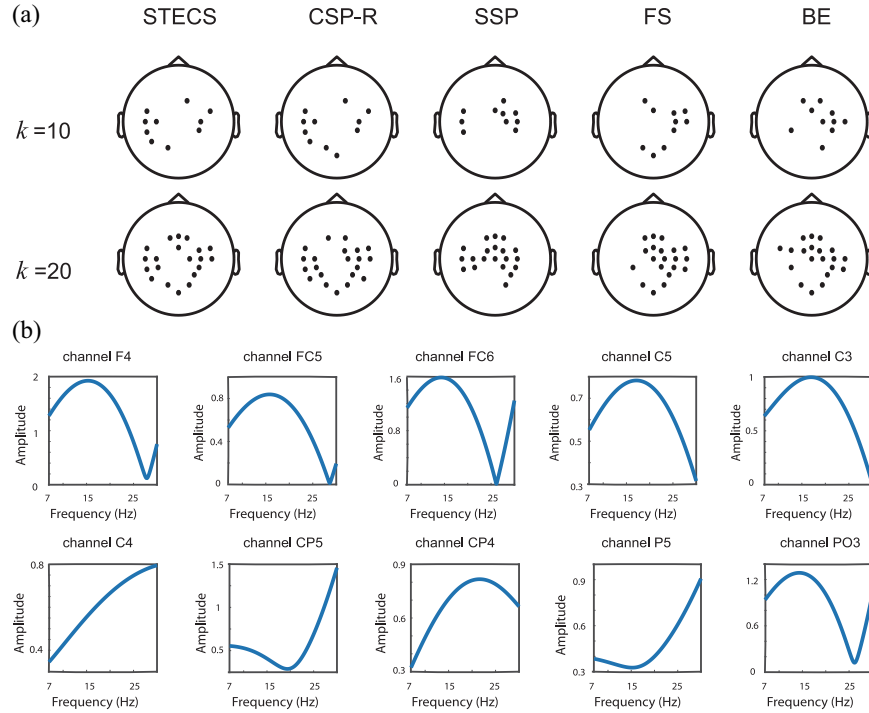


Fig. 5. (a) Topographical maps of the selected channels for subject S7, for which the performing tasks were right-hand and foot motor imagery. (b) Frequency response curves for the optimal temporal filters from subject S7, and the ten channels are selected when $k = 10$.

respectively. The results are more conspicuous when the number of selected channels is relatively small. The frequency response is the largest in the alpha and beta bands, which is also consistent with prior neurophysiological findings. In comparison, the channels selected by FS for *aw* are mainly concentrated near the temporal regions, which are implausible from a neurophysiological viewpoint. Moreover, several channels selected by SSP for *K3-23* when $k = 10$ are in the prefrontal regions, which are not supposed to significantly contribute to ERD. These problematic patterns can explain the poorer performance of the FS and SSP algorithms. Notably, the selected channels when k is small are not necessarily a subset of those when k is larger due to the multivariate nature of STECS.

Next, in order to assess how often certain channels are selected across subjects for STECS, the times of each channel being selected across all the data subsets within each dataset is shown in Fig. 6 for different k 's. It can be seen that a predominant number of the selected channels is centered at the sensorimotor regions associated with the hand and foot motor imagery, which is reasonable from a neurophysiological view.

D. Computational Cost

In STECS, channel selection via solving problem (9) is based on L-BFGS, which can be implemented efficiently. Hence, STECS can converge quickly. For instance, on a Windows PC with a 3.20-GHz Inter Core i5-3470 CPU and 8-GB RAM, in MATLAB 7.7, the runtimes of STECS applied to subjects *aw*, *K3-23*, and *S7*, with the regularization parameter ρ_1 covering all the candidate values $B = \{0.01, 0.02, \dots, 0.7\}$, are 518.34 s, 67.18 s, and 20.2 s

for subjects *aw*, *K3-23*, and *S7*, respectively. That is, the average runtimes for each ρ_1 are 7.40 s, 0.96 s, and 0.29 s when the numbers of the recorded channels are 118, 60, and 32, respectively. This suggests that STECS holds the promise for applications to quasi-real-time BCI systems.

IV. DISCUSSION

A. Connections With Existing Methods

Several existing algorithms can be viewed as special cases of STECS: 1) when the group sparsity constraint parameter $\rho_1 = 0$, that is, the EEG signals are analyzed using full-cap channels, STECS is reduced to STRCSP; 2) setting $N = 1$, $\rho_1 = 0$, and $\rho_2 = 0$ in STECS yields CSP; and 3) SSP can be obtained by setting the temporal filter order $N = 1$ and the regularization parameter $\rho_2 = 0$ in STECS.

To better reveal connections, according to the spatiotemporal information leveraged for channel selection (spatiotemporal filtering, spatial filtering, or no channel selection) and feature extraction (spatiotemporal filtering or spatial filtering), the existing algorithms are summarized in Table IV.

B. Temporal Filtering Helps Channel Selection

To specifically assess the impact of temporal filtering on channel selection, we remove temporal filtering from the channel selection stage of STECS, which leads to SSP + STRCSP (called SSP-R henceforth), where SSP is employed for channel selection, and STRCSP is then applied to the selected channels to optimize multiple spatiotemporal filters. In line with STECS, the order of the temporal filter and regularization

TABLE V
MEAN CLASSIFICATION PERFORMANCE (%) OF STECS AND SSP-R ON THE THREE DATASETS

—	$k = 10$	$k = 20$	$k = 30$	$k = 40$
SSP-R	84.05 ± 12.84 ($p = 0.3682$)	86.06 ± 13.73 ($p = 0.1737$)	87.16 ± 13.19 ($p = 0.0677$)	87.18 ± 13.08 ($p = 0.1230$)
STECS	85.98 ± 10.93	88.49 ± 11.24	89.68 ± 10.11	89.89 ± 10.36

TABLE VI
MEAN CLASSIFICATION PERFORMANCE (%) OF STECS AND C-STECS ON THE THREE DATASETS

—	$k = 10$	$k = 20$	$k = 30$	$k = 40$
C-STECS	84.96 ± 13.90 ($p = 0.3140$)	86.57 ± 12.81 ($p = 0.0611$)	86.62 ± 12.44 ($p = 0.0048$)	85.86 ± 12.76 ($p = 0.0160$)
STECS	85.98 ± 10.93	88.49 ± 11.24	89.68 ± 10.11	89.89 ± 10.36

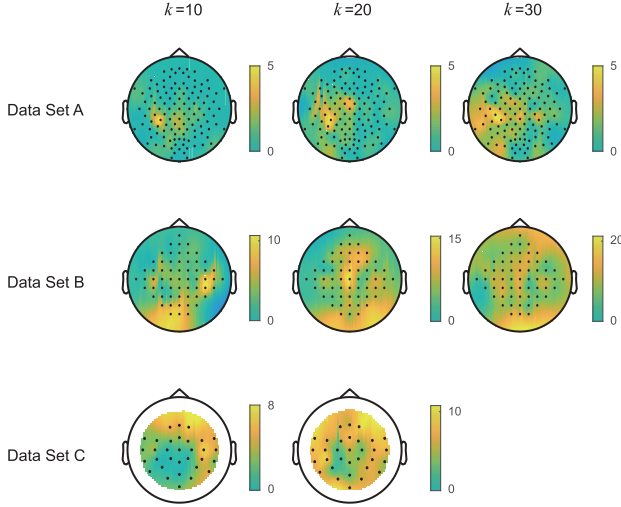


Fig. 6. Visualization of selected channels across subjects. Each column corresponds to a different k . Colors in each topographical map encode the times of each channel being selected across all the data subsets within each dataset.

parameter are set to $N = 6$, and $\rho_2 = 10^{-5}$. The classification accuracies of SSP-R and STECS on the three datasets are displayed in Table V, where the p values by the Wilcoxon signed-rank test are presented in brackets.

The mean classification accuracies of STECS are 1.93%, 2.43%, 2.52%, and 2.71% higher than SSP-R when $k = 10, 20, 30$, and 40 , respectively. The performance of STECS on the data subsets is better than that of SSP-R, even though the differences are not significant. Therefore, incorporating spatiotemporal information for channel selection is recommended for practical application.

C. Channel-Specific Versus Channel-Invariant Temporal Filtering for Channel Selection

STECS assumes that discriminative information may be contained in different frequency bands for different channels, which is supported by prior neurophysiological studies [38]. To demonstrate that optimizing channel-specific temporal filters is more advantageous to classification than optimizing an identical temporal filter, we compare STECS with its variant, called constrained STECS (C-STECS), in which a single temporal filter is shared across all channels in both the channel selection and spatiotemporal filter optimization stages.

The objective function for C-STECS is as follows:

$$\min_{\tilde{\mathbf{w}}} J_6(\tilde{\mathbf{w}}) \triangleq \tilde{\mathbf{w}}^\top \tilde{\mathbf{R}}_2 \tilde{\mathbf{w}} + \frac{1}{\tilde{\mathbf{w}}^\top \tilde{\mathbf{R}}_1 \tilde{\mathbf{w}}} + \rho_3 \cdot \|\tilde{\mathbf{w}}\|_{2,1} \quad (12)$$

where $\tilde{\mathbf{a}} = [\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_N]^\top \in \mathbb{R}^{N \times 1}$ denotes the temporal filter shared by all the C channels, and $\tilde{\mathbf{w}} = \tilde{\mathbf{a}} \otimes \mathbf{s} = [\tilde{a}_1 \mathbf{s}^\top, \tilde{a}_2 \mathbf{s}^\top, \dots, \tilde{a}_N \mathbf{s}^\top]^\top \in \mathbb{R}^{NC \times 1}$ is the reparametrized spatiotemporal filter.

Details on the derivation and optimization algorithm for C-STECS are provided in Appendix D. Table VI shows the mean classification accuracies of STECS and C-STECS on the three datasets when $k = 10, 20, 30$, and 40 , and the mean classification accuracies of STECS are higher by 1.02%, 1.92%, 3.06%, and 4.03% over C-STECS, respectively. The Bonferroni-corrected Wilcoxon signed rank test indicates that, STECS significantly outperforms C-STECS when $k = 30$ and 40 ($p = 0.0048$ for $k = 30$ and $p = 0.0160$ for $k = 40$). This result provides evidence for the superiority of optimizing channel-specific temporal filters over optimizing a channel-invariant temporal filter.

D. Limitations

Despite the highly encouraging performance of STECS for EEG channel selection on three motor imagery BCI datasets, this article can be improved along the following lines. First, more EEG datasets are required to assess STECS and establish its superiority over other channel selection approaches. Second, in this article, the hyperparameters of STECS are empirically determined based on our previous analysis. More principled ways of hyperparameter optimization such as grid search with cross-validation will be employed in our future work to address this issue. Third, in STECS, channel selection, optimization of multiple spatiotemporal filters, and classification are carried out in separate stages with distinct optimization criteria. An integrated framework for simultaneous channel selection, spatiotemporal filter optimization, and classification under a unified criterion may further enhance the classification performance.

V. CONCLUSION

This article proposed a new channel selection method called STECS based on exploiting the underlying spatiotemporal information of EEG. STECS is applied to three motor imagery EEG datasets, in which the numbers of recorded channels are 118, 60, and 32, respectively. The results demonstrate

that STECS significantly outperforms the existing CSP-related channel selection algorithms. Moreover, there is no significant performance decline compared with the state-of-the-art full-cap-based spatiotemporal filtering methods, and the selected channels are neurophysiologically meaningful, suggesting the effectiveness of STECS. The algorithm can also be readily deployed for quasi-real-time BCI experiments due to its low computational complexity, and therefore, holds the promise of enhancing the practical utility of high-performance BCIs.

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